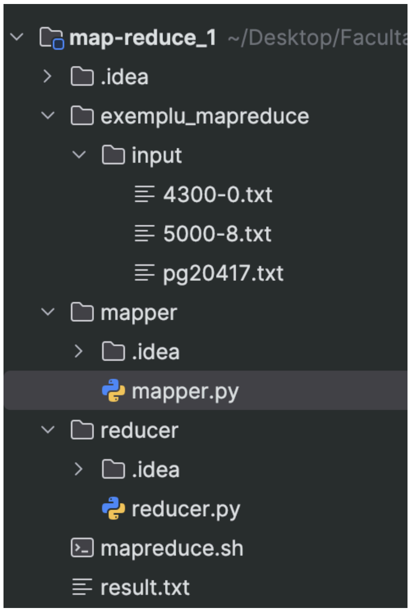
Să se implementeze un algoritm MapReduce care, pentru un set de fișiere text, va grupa cuvintele în funcție de **prima literă** a fiecărui cuvânt.



NOTITE

1. Instalezi pachetele care trebuie din stanga jos
2. Creezi dupa structura din poza fisierele result si cele din input
3. Rulezi din consola ./[mapreduce.sh](). Daca nu se poate rula ii dai permisiuni: chmod +x mapreduce.sh
4. Verifici in result ca ai structiu “Litera {Cuvinte care incep cu litera}”

[mapper.py]()

```python
#!/usr/bin/env python
import sys

# input comes from STDIN (standard input)
for line in sys.stdin:
   # remove leading and trailing whitespace
   line = line.strip()
   # split the line into words
   words = line.split(" ")
   # increase counters
   for word in words:
       # write the results to STDOUT (standard output);
       # what we output here will be the input for the
       # Reduce step, i.e. the input for reducer.py
       #
       # tab-delimited; the trivial word count is 1
       word = ''.join(e for e in word if e.isalnum())
```

```
        if word != '':
            print('{}\t{}'.format(word[0], word))
```

reduccer.py

```
#!/usr/bin/env python
"""reducer.py"""
import re
import sys

words_dictionary = {}

# input comes from STDIN
for line in sys.stdin:
   # remove leading and trailing whitespace
   line = line.strip()
   re.split("['/ ]", line)

   # parse the input we got from mapper.py
   letter, word = line.split()  # line.split('\t', 1)

   if letter in words_dictionary:
       if word not in words_dictionary[letter]:
           words_dictionary[letter].append(word)
   else:
       words_dictionary[letter] = list(word)

# do not forget to output the last word if needed!
for key in words_dictionary:
   print('{}\t{}'.format(key, words_dictionary[key]))
```

mapreduce.sh

```
rm result.txt
cat ./exemplu_mapreduce/input/4300-0.txt ./exemplu_mapreduce/input/pg20417.txt
| python3 ./mapper/mapper.py | python3 ./reducer/reducer.py > result.txt
```